

Profile

AI/ML Engineer with 1+ years of professional experience building production-grade **Computer Vision, NLP, and Generative AI systems**. Skilled in designing end-to-end pipelines spanning **LLMs, RAG, and Vision-Language Models** on AWS/GCP, with a track record of reducing inference costs and infrastructure spend while improving accuracy and reliability at scale.

Technical Skills

AI & ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, YOLO, Vision-Language Models (VLMs), Fine-tuning (LoRA / QLoRA), Stable Diffusion, CNNs, Medical Image Analysis

MLOps & Infra: AWS (EC2, S3), Google Cloud Platform (GCP), Docker, CI/CD Pipelines, REST APIs, Vector Databases (FAISS, ChromaDB)

Frameworks & Libraries: PyTorch, LangChain, OpenCV, Scikit-learn, HuggingFace Transformers, Streamlit, Django, Flask, React.js

Languages: Python, C++, SQL, Linux, R

Professional Experience

Python AI Developer

Feb 2026 – Present

Ayant Technologies

Jaipur, India

- Architected a hybrid Computer Vision pipeline combining YOLO object detection with Vision-Language Models (VLMs) for real-time security monitoring, achieving an **85% reduction** in LLM API costs and a **40% reduction** in AWS infrastructure spend.
- Built a production-grade face recognition and tracking system powered by FAISS vector indexing for sub-millisecond similarity search, delivering **85% identification accuracy** at enterprise scale.
- Designed and maintained end-to-end CI/CD pipelines on AWS and GCP, automating model deployment and feature integration into core production systems with **zero-downtime** releases.

AI Engineer

June 2025 – Jan 2026

NAVOLDEN STONE PRIVATE Limited

Jaipur, India

- Engineered a **Retrieval-Augmented Generation (RAG) pipeline** for customer FAQs, implementing document preprocessing, semantic embeddings, vector search, and LLM-based response generation for context-aware customer interactions.
- Developed a **personalized recommendation system** that modeled customer personas, preferences, and gemstone attributes to generate ranked product recommendations tailored to individual users.
- Integrated AI services into web applications through **Python REST APIs**, optimizing retrieval, recommendation, and inference workflows for reliable real-world deployment.

Projects

MedicoX | Python, CNN, Django, React.js, Deep Learning

Dec 2024 – Mar 2025

- Engineered three independent CNN-based medical image analysis models for detecting Alzheimer's disease (MRI), Lung Cancer (CT scan), and Pneumonia (X-ray), achieving clinically relevant classification accuracy across all modalities.
- Designed a multi-class classification architecture capable of correctly stratifying diseases into severity/type subcategories, enabling actionable diagnostic insights from raw imaging data.
- Integrated models into a full-stack web application using Django REST Framework as the backend API layer and React.js as the interactive frontend, delivering a seamless end-to-end diagnostic tool.

PDF Chatbot | Django, LangChain, Groq, LLaMA 3, ChromaDB, Docker

Apr 2025 – May 2025

- Built a document-aware conversational AI agent leveraging LLaMA 3 and Pytesseract OCR to enable semantic querying over unstructured PDFs, reducing manual document search time by **90%**.
- Integrated ChromaDB for high-speed vector search with custom HuggingFace embeddings; containerized the full application via Docker for reproducible, environment-agnostic deployment.

Narration Generation Artwork | Stable Diffusion, TTS, Django, ML

Jul 2025 – Aug 2025

- Built a multimodal generative AI platform that transforms text prompts into narrated audio stories, combining Microsoft Edge TTS for speech synthesis with dynamically mixed background music for immersive storytelling.
- Integrated the Stability AI API (Stable Diffusion) to auto-generate high-fidelity visuals from prompts, delivering a fully automated text → speech → music → image pipeline for multimedia content creation.

Education

Swami Keshvanand Institute of Technology (SKIT), Jaipur

2021 – 2025

B.Tech in Data Science Engineering

CGPA: 8.4 / 10.0

All India Senior Secondary School Examination

2018 – 2020

Intermediate (PCM)

Percentage: 85%